



Student Number	
Mark / 44	

Chemistry

HSC Course

Acidic Environment + Industrial Chemistry

Theory Test • 2007

General Instructions

- Reading time – 5 minutes
- Working time – 45 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A Data Sheet and a Periodic Table are provided
- Write your Student Number at the top of this page

Total Marks – 44

Part A – 12 marks

- Attempt Questions 1 – 12
- Allow about 10 minutes for this part

Part B – 32 marks

- Attempt Questions 13 – 20
- Allow about 40 minutes for this part

Allow about 10 minutes for this part

A  B  C  D 

correct (arrow pointing to B)

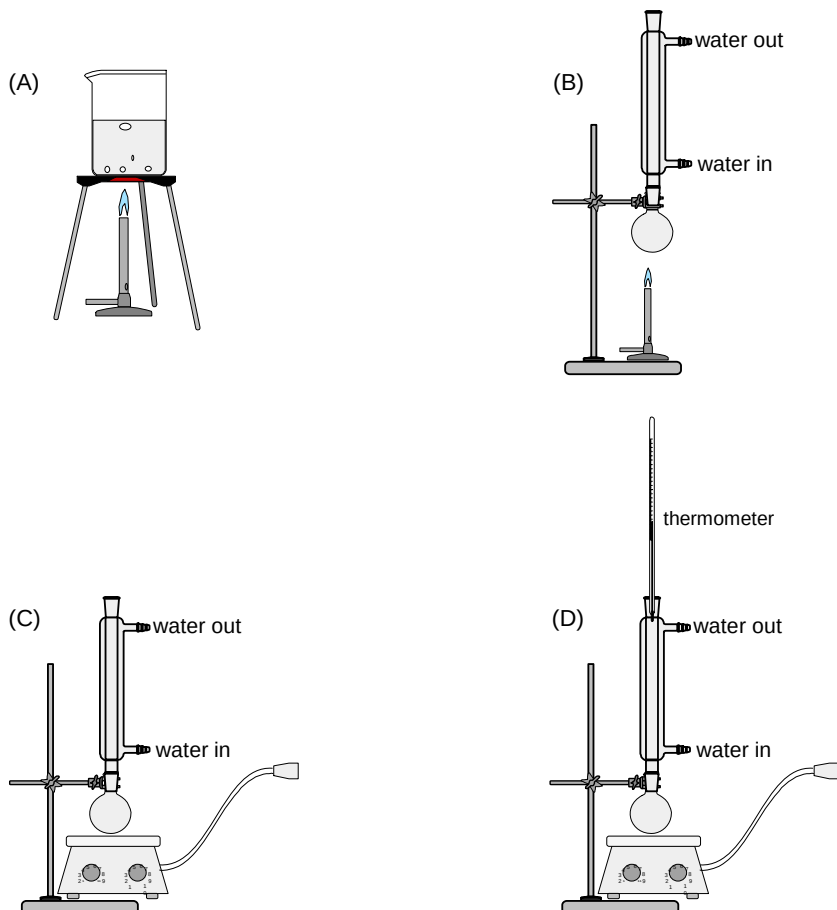
Answer Box for Questions 1 – 12				
1	A ☐	B ☐	C ☐	D ☐
2	A ☐	B ☐	C ☐	D ☐
3	A ☐	B ☐	C ☐	D ☐
4	A ☐	B ☐	C ☐	D ☐
5	A ☐	B ☐	C ☐	D ☐
6	A ☐	B ☐	C ☐	D ☐
7	A ☐	B ☐	C ☐	D ☐
8	A ☐	B ☐	C ☐	D ☐
9	A ☐	B ☐	C ☐	D ☐
10	A ☐	B ☐	C ☐	D ☐
11	A ☐	B ☐	C ☐	D ☐
12	A ☐	B ☐	C ☐	D ☐

- 1 Which of the following substances is a natural source of atmospheric NO_2 ?
- (A) combustion of coal
 - (B) combustion of unleaded petrol
 - (C) lightning storm
 - (D) volcanoes
- 2 Which of the following is the correct IUPAC name for citric acid?
- (A) 2-hydroxypropanoic acid
 - (B) 1,2,3-tripropanoic acid
 - (C) hydroxypropane carboxylic acid
 - (D) 2-hydroxypropane-1,2,3-tricarboxylic acid
- 3 Which of the following would be the most appropriate chemical to neutralise a small amount of acid spilt on a lab benchtop.
- (A) CH_3COOH
 - (B) H_2O
 - (C) NaOH
 - (D) NaHCO_3
- 4 A pigment from violet flowers is purple in acidic solutions, yellow in alkaline solutions and colourless in neutral solutions. In which of the following would you expect the pigment to show no colour?
- (A) a solution of sodium chloride
 - (B) white vinegar
 - (C) household ammonia (cleaning agent)
 - (D) lemonade (colourless lemonade)
- 5 Which of the following substance will have the lowest boiling point?
- (A) HCOOH
 - (B) CH_3OH
 - (C) H_2O
 - (D) CH_4

6 Identify the main product of the reaction between $\text{CH}_3(\text{CH}_2)_4\text{COOH}$ and $\text{CH}_3\text{CH}_2\text{OH}$.

- (A) pentyl ethanoate
- (B) ethyl butanoate
- (C) hexyl ethanoate
- (D) ethyl hexanoate

7 Which of the set-ups below is most appropriate to produce esters in the laboratory?



8 Which of the following best describes a 2.0 mol L^{-1} solution of acetic acid?

- (A) a concentrated solution of a weak acid
- (B) a weak solution of a dilute acid
- (C) a strong solution of a concentrated acid
- (D) a dilute solution of a strong acid

9 Which of the following chemists proposed that all acids contain hydrogen?

- (A) Arrhenius
- (B) Brønsted
- (C) Davy
- (D) Lavoisier

10 What is the pH of $3.6 \times 10^{-3} \text{ mol L}^{-1} \text{ HNO}_3$?

- (A) 0.0036
- (B) 2.44
- (C) 3.60
- (D) 5.63

11 The table shows the colour ranges of several indicators.

<i>Indicator</i>	<i>low range colour</i>	<i>pH of transition range</i>	<i>high range colour</i>
thymol blue	red	1.2 – 2.8	yellow
bromophenol blue	yellow	3.0 – 4.6	blue
methyl orange	red	3.1 – 4.4	yellow
methyl red	red	4.8 – 6.0	yellow
litmus	red	4.7 – 8.2	blue
thymol blue	yellow	8.0 – 9.6	blue
phenolphthalein	colourless	8.3 – 10.0	red

The results of testing a substance with three of the indicators are shown in the table.

<i>Indicator added to substance</i>	<i>Result (final colour of indicator)</i>
methyl orange	red
thymol blue	yellow
phenolphthalein	colourless

Which is a possible pH of the substance?

- (A) 1
- (B) 3
- (C) 4.5
- (D) 8

12 What is the purpose of concentrated sulfuric acid in the esterification reaction?

- (A) To speed up the attainment of equilibrium
- (B) To shift the equilibrium to the left
- (C) To increase the yield of ester
- (D) To increase the activation energy of the reaction

Part B – 32 marks

Attempt Questions 13 – 20

Allow about 40 minutes for this part

► *Show all relevant working in questions involving calculations.*

Question 13 (5 marks)

Cloudy ammonia (a household cleansing product) is a solution of ammonia in water which is excellent for cleaning glass. The Amalgamated Ammonia Company regularly tests the concentration of its manufactured cloudy ammonia by a simple acid–base titration using standardised sulfuric acid.

- (a) Write the balanced chemical equation for the neutralisation of cloudy ammonia by sulfuric acid. **(1 mark)**

- (b) Suggest a suitable indicator for this acid–base titration and justify your answer. **(2 marks)**

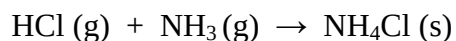
- (c) A lab technician analyses a sample of cloudy ammonia from the production line. The table shows the titration data.

<i>Concentration of standardised H_2SO_4</i>	0.155 mol L ⁻¹
<i>Average titre of standardised H_2SO_4</i>	47.4 mL
<i>Aliquot of cloudy ammonia</i>	5.00 mL

Calculate the concentration of cloudy ammonia. **(2 marks)**

Question 14 (3 marks)

The reaction of hydrogen chloride gas with ammonia gas,



is classified as an acid /base reaction according to Brønsted/Lowry theory, but not according to Arrhenius' theory of acids and bases.

Justify this statement.

Question 15 (5 marks)

- (a) Identify the following salts as acidic, basic or neutral. **(2 marks)**

<i>Salt</i>	<i>Acidic, basic or neutral solution</i>
NaCl	
NH ₄ Cl	
NaHCO ₃	
KCH ₃ COO	

- (b) Write an equation to show the action of an acidic salt in (a), in water. **(1 mark)**

- (c) Identify one conjugate acid/base pair in the reaction given in (b) above. **(1 mark)**

- (d) Identify an amphoteric ion present in one of the salts listed in the table. **(1 mark)**

Question 16 (7 marks)

Atmospheric oxides of sulfur have been increasing over the last 150 years.

- (a) Assess the impact of this trend and include relevant balanced equations in your answer. **(5 marks)**
 ► Do not reproduce the reaction from part (b) below.

- (b) Copper may be extracted from copper(II) sulfide ore by roasting it in oxygen. Copper(II) oxide and sulfur dioxide are produced.

What volume of sulfur dioxide is produced when 191 g of copper(II) sulfide is roasted? (Volume measured at 25°C and 100 kPa). **(2 marks)**

Question 17 (3 marks)

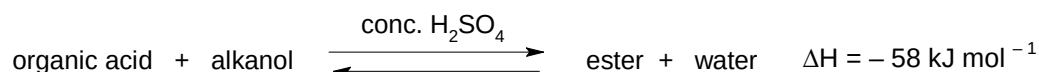
A primary standard solution was prepared by dissolving 0.316 g of hydrated oxalic acid, $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$ in water and making the solution up to 250.0 mL.

- (a) Calculate the concentration of the solution. **(2 marks)**

- (b) Identify the piece of accurate glassware in which this solution is prepared. **(1 mark)**

Question 18 (2 marks)

An organic acid and an alcohol were allowed to react using the appropriate apparatus, producing an ester and water.



Certain changes are then made to this mixture. Considering each separately, state the effect that the change has on the equilibrium position by completing the table.

<i>Change/Stress</i>	<i>Effect on the equilibrium position</i>
(a) raising the temperature	
(b) increasing the pressure above the solution	
(c) adding alkanol to the mixture	
(d) adding a base	

Question 19 (3 marks)

Selected students from Hiuai High School were assigned to plan and perform an experiment to decarbonate a cola drink. Detailed below are the steps they took to address the problem.

- Placed a pre-weighed 400 mL beaker on a warm hot plate.
- Unscrewed a 375 mL bottle of JR cola and carefully transferred the contents to the beaker.
- The beaker was weighed immediately after transfer and weighed every 2 minutes thereafter.
- At intervals between weighings, the beaker was kept on the hot plate and stirred with a thermometer to monitor the temperature which was kept at 60°C for 10 minutes.

Assess the validity of the procedure.

Question 20 (4 marks)

Identify a non-fossil fuel natural product and the issues associated with the need to replace this natural product. Assess the viability of a possible replacement with respect to the issues.



**Answers
and
Marking Scheme**

Chemistry

HSC Course

Acidic Environment + Industrial Chemistry

Theory Test • 2007

Allow about 10 minutes for this part

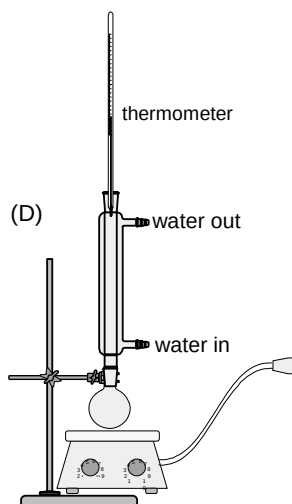
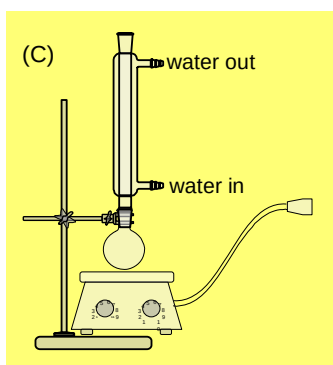
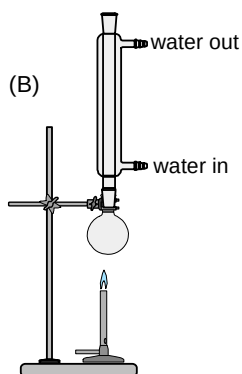
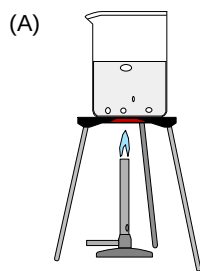
A  B  C  D 

correct (with an arrow pointing to B)

Answer Box for Questions 1 – 12				
1	A ☐	B ☐	C ☒	D ☐
2	A ☐	B ☐	C ☐	D ☒
3	A ☐	B ☐	C ☐	D ☒
4	A ☒	B ☐	C ☐	D ☐
5	A ☐	B ☐	C ☐	D ☒
6	A ☐	B ☐	C ☐	D ☒
7	A ☐	B ☐	C ☒	D ☐
8	A ☒	B ☐	C ☐	D ☐
9	A ☐	B ☐	C ☒	D ☐
10	A ☐	B ☒	C ☐	D ☐
11	A ☐	B ☒	C ☐	D ☐
12	A ☒	B ☐	C ☐	D ☐

- 1 Which of the following substances is a natural source of atmospheric NO_2 ?
- (A) combustion of coal
 - (B) combustion of unleaded petrol
 - (C) lightning storm
 - (D) volcanoes
- 2 Which of the following is the correct IUPAC name for citric acid?
- (A) 2-hydroxypropanoic acid
 - (B) 1,2,3-tripropanoic acid
 - (C) hydroxypropane carboxylic acid
 - (D) 2-hydroxypropane-1,2,3-tricarboxylic acid
- 3 Which of the following would be the most appropriate chemical to neutralise a small amount of acid spilt on a lab benchtop.
- (A) CH_3COOH
 - (B) H_2O
 - (C) NaOH
 - (D) NaHCO_3
- 4 A pigment from violet flowers is purple in acidic solutions, yellow in alkaline solutions and colourless in neutral solutions. In which of the following would you expect the pigment to show no colour?
- (A) a solution of sodium chloride
 - (B) white vinegar
 - (C) household ammonia (cleaning agent)
 - (D) lemonade (colourless lemonade)
- 5 Which of the following substance will have the lowest boiling point?
- (A) HCOOH
 - (B) CH_3OH
 - (C) H_2O
 - (D) CH_4
- 6 Identify the main product of the reaction between $\text{CH}_3(\text{CH}_2)_4\text{COOH}$ and $\text{CH}_3\text{CH}_2\text{OH}$.
- (A) pentyl ethanoate
 - (B) ethyl butanoate
 - (C) hexyl ethanoate
 - (D) ethyl hexanoate

7 Which of the set-ups below is most appropriate to produce esters in the laboratory?



8 Which of the following best describes a 2.0 mol L^{-1} solution of acetic acid?

- (A) a concentrated solution of a weak acid
- (B) a weak solution of a dilute acid
- (C) a strong solution of a concentrated acid
- (D) a dilute solution of a strong acid

9 Which of the following chemists proposed that all acids contain hydrogen?

- (A) Arrhenius
- (B) Brønsted
- (C) Davy
- (D) Lavoisier

10 What is the pH of $3.6 \times 10^{-3} \text{ mol L}^{-1} \text{ HNO}_3$?

- (A) 0.0036
- (B) 2.44
- (C) 3.60
- (D) 5.63

- 11 The table shows the colour ranges of several indicators.

Indicator	low range colour	pH of transition range	high range colour
thymol blue	red	1.2 – 2.8	yellow
bromophenol blue	yellow	3.0 – 4.6	blue
methyl orange	red	3.1 – 4.4	yellow
methyl red	red	4.8 – 6.0	yellow
litmus	red	4.7 – 8.2	blue
thymol blue	yellow	8.0 – 9.6	blue
phenolphthalein	colourless	8.3 – 10.0	red

The results of testing a substance with three of the indicators are shown in the table.

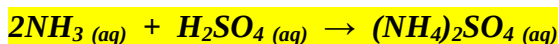
<i>Indicator added to substance</i>	<i>Result (final colour of indicator)</i>
methyl orange	red
thymol blue	yellow
phenolphthalein	colourless

Which is a possible pH of the substance?

- (A) 1
(B) 3
(C) 4.5
(D) 8
- 12 What is the purpose of concentrated sulfuric acid in the esterification reaction?
- (A) To speed up the attainment of equilibrium
(B) To shift the equilibrium to the left
(C) To increase the yield of ester
(D) To increase the activation energy of the reaction

- 13 Cloudy ammonia (a household cleansing product) is a solution of ammonia in water which is excellent for cleaning glass. The Amalgamated Ammonia Company regularly tests the concentration of its manufactured cloudy ammonia by a simple acid–base titration using standardised sulfuric acid.

- (a) Write the balanced chemical equation for the neutralisation of cloudy ammonia by sulfuric acid. (1 mark)



► $\text{NH}_3(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$ was also accepted

- (b) Suggest a suitable indicator for this acid–base titration and justify your answer. (2 marks)

Methyl orange (3.1 – 4.4), methyl red (4.4 – 6.2), bromophenol blue (3 – 4.6) would be suitable indicators, since the pH at the equivalence point is acidic, a result of the acidic nature of the salt formed, $(\text{NH}_4)_2\text{SO}_4$.

► Bromothymol blue (6 – 7.6) not acceptable due to a large end point error.

► Correct indicator stated = 1 mark

► “... acidic salt formed ...” = 1 mark

- (c) A lab technician analyses a sample of cloudy ammonia from the production line. The table shows the titration data.

Concentration of standardised H_2SO_4	0.155 mol L ⁻¹
Average titre of standardised H_2SO_4	47.4 mL
Aliquot of cloudy ammonia	5.00 mL

Calculate the concentration of cloudy ammonia. (2 marks)

$$\text{mol } \text{H}_2\text{SO}_4 = c V = 0.155 \text{ mol L}^{-1} \times 0.0474 \text{ L} = 7.347 \times 10^{-3}$$

$$\text{mol } \text{NH}_3 = 2 \times \text{mol } \text{H}_2\text{SO}_4 = 1.469 \times 10^{-2}$$

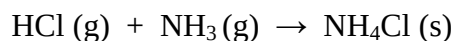
$$[\text{NH}_3] = \text{mol} \div V = 1.469 \times 10^{-2} \div 0.00500 \text{ L} = 2.9388 = 2.94 \text{ mol L}^{-1}$$

► Correct calculation of moles of NH_3 = 1 mark

► Divide by 0.00500 L = 1 mark

► Flow on error from student’s equation provided in (a) was accepted if the equation provided was incorrect.

- 14 The reaction of hydrogen chloride gas with ammonia gas,



is classified as an acid /base reaction according to Brønsted/Lowry theory, but not according to Arrhenius' theory of acids and bases.

Justify this statement. (3 marks)

Sample Answer

Arrhenius proposed that acids are substances that produce hydrogen ions in water and bases were substances that dissociated in water to produce hydroxide ions. This would not be an Arrhenius acid/base reaction because no hydroxide ions are produced. Brønsted/Lowry defined acids as proton donors and bases as proton acceptors.

In this case, HCl donates a proton and NH₃ accepts a proton so is an acid/base reaction.

Marking criteria	Marks
Describes both acid theories and explains why this is an acid/base reaction.	3
Describes Brønsted/Lowry theory and explains why this is an acid /base reaction	2
Correctly defines acids according to Arrhenius or Brønsted/Lowry	1

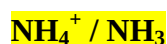
- 15 (a) Identify the following salts as acidic, basic or neutral. (2 marks)

Salt	Acidic, basic or neutral solution
NaCl	<i>neutral</i>
NH ₄ Cl	<i>acidic</i>
NaHCO ₃	<i>basic</i>
KCH ₃ COO	<i>basic</i>

- (b) Give an equation to show the action of an acidic salt in (a), in water. (1 mark)



- (c) Identify one conjugate acid/base pair in the reaction given in (b) above. (1 mark)



- (d) Identify an amphiprotic ion present in one of the salts listed in the table. (1 mark)



16 Atmospheric oxides of sulfur have been increasing over the last 150 years.

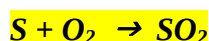
(a) Assess the impact of this trend and include relevant balanced equations in your answer. (5 marks) ► Do not reproduce the reaction from part (b) below.

(b) Copper may be extracted from copper(II) sulfide ore by roasting it in oxygen. Copper(II) oxide and sulfur dioxide are produced.

What volume of sulfur dioxide is produced when 191 g of copper(II) sulfide is roasted? (Volume measured at 25°C and 100 kPa). (2 marks)

Sample answer

(a) *There has been a rapid increase in atmospheric SO₂ over the last 150 years due to the industrial revolution which caused the increased burning of coal and the release of SO₂ from metal smelting.*



The consequence of an increase on the SO₂ has been an environmental problem called acid rain.



This acid rain has caused widespread deforestation and damage to buildings and waterways. The impact has been great and has led to legislation to minimize the output of SO₂ into the atmosphere by industry.

Marking criteria	Marks
All of 4 and a judgement	5
One or two equations and a description of acid rain and its consequences	3 – 4
Two equations or one equation and acid rain	2
One equation or acid rain	1

(b) *mol CuS = 2.00 mol* (1 mark)

V SO₂ = 49.5 L (1 mark)

- 17 A primary standard solution was prepared by dissolving 0.316 g of hydrated oxalic acid, $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$ in water and making the solution up to 250.0 mL

(a) Calculate the concentration of the solution. (2 marks)

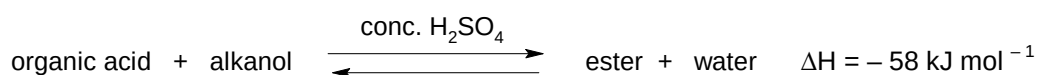
$$\text{mol } (\text{COOH})_2 \cdot 2\text{H}_2\text{O} = 2.5066 \times 10^{-3}$$

$$[(\text{COOH})_2 \cdot 2\text{H}_2\text{O}] = 0.0100 \text{ mol L}^{-1}$$

(b) Identify the piece of accurate glassware in which this solution is prepared. (1 mark)

volumetric flask

- 18 An organic acid and an alcohol were allowed to react using the appropriate apparatus, producing an ester and water.



Certain changes are then made to this mixture. Considering each separately, state the effect that the change has on the equilibrium position by completing the table. (2 marks)

<i>Change/Stress</i>	<i>Effect on the equilibrium position</i>
(a) raising the temperature	shifts to the left
(b) increasing the pressure above the solution	no effect
(c) adding alkanol to the mixture	shifts to the right
(d) adding a base	shifts to the left

- 19 Selected students from Hiuai High School were assigned to plan and perform an experiment to decarbonate a cola drink. Detailed below are the steps they took to address the problem.

- Placed a pre-weighed 400 mL beaker on a warm hot plate.
- Unscrewed a 375 mL bottle of JR cola and carefully transferred the contents to the beaker.
- The beaker was weighed immediately after transfer and weighed every 2 minutes thereafter.
- At intervals between weighings, the beaker was kept on the hot plate and stirred with a thermometer to monitor the temperature which was kept at 60°C for 10 minutes.

Assess the validity of the procedure. (3 marks)

Criteria	Mark
2 mistakes in the procedure	2
final assessment (judgement)	1

Possible Answer

The following mistakes were committed by the students making their procedure invalid:

- the original mass of the cola was not recorded
- the cola was heated resulting in possible loss of water in addition to the loss of CO₂
- the method followed would not allow the sample to be weighed to constant mass

- 20 Identify a non-fossil fuel natural product and the issues associated with the need to replace this natural product. Assess the viability of a possible replacement with respect to the issues. (4 marks)

Criteria	Mark
Identification of a non-fossil fuel natural product and replacement	2
Viability of the replacement with respect to the issues (for and against)	2

Possible Answer

- A natural product that may require replacement is the anti-malarial drug, Quinine.
- Quinine is extracted from the bark of the cinchona tree, which remains to be the only viable source of Quinine. Synthetic quinine was produced in the 1940s but this could not compete in economic terms with that produced naturally.
- The supply of quinine, was limited by the availability of the cinchona tree and a synthetic replacement was required.
- Chloroquine is a synthetic drug which can be produced in bulk to replace quinine in the treatment of malaria.
- However, in recent years increasing resistance to chloroquine has been developed by certain strains of plasmodium, the mosquito parasite which causes malaria.
- Natural quinine extracted from quinine bark and the use of natural bark tea and/or bark extracts are making a comeback in the management and treatment of malaria.